

2. DECISION MAKING

SOME DM-PROBLEMS



WHICH CHOICE DO YOU PREFER?

- Choice A: You take a bet in which a fair coin is tossed.
 - If the toss comes up heads, you win \$10.
 - If the toss comes up tails, you lose \$10.
- Choice B: You don't take the bet.

YOU HAVE 5 SECONDS!!!!

- Estimate the value of the following product:
- $1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8$

DISEASE Y PROBLEM

If a test, to detect a disease Y whose prevalence is 1/1000, has a false positive rate of 5%, what is the chance that a person found to have a positive result actually has the disease, assuming that you know nothing about the person's symptoms or signs? ----%

LINDA-PROBLEM

- Linda is 31 years old, single, outspoken, and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice, and also participated in anti-nuclear demonstrations.
- Please rank the following statements by their probability, using 1 for the most probable and 3 for the least probable.
 - a. Linda is a teacher in an elementary school.
 - b. Linda is a bank teller.
 - c. Linda is a bank teller and is active in the feminist movement.

IN AN EPIDEMIC, WHICH OF THE FOLLOWING TWO CHOICES DO YOU PREFER?

- CHOICE 1:
 - 80% chance of losing 100 lives
- CHOICE 2:
 - 100% chance of losing 75 lives

WHICH TYPE OF FOUR-LETTER WORD HAS MORE MEMBERS?

- Words that begin with an R:
 - R _ _ _
- Words that have R in the third position.:
 - _ _ R _

LAWYERS VS ENGINEERS: INSTRUCTIONS

- Divide the class in 2 groups of roughly the same size
- While one group is reading the assignment (in silence), the other one keeps their eyes closed.
- On my signal, we switch roles.
- Afterwards, you all individually solve 2 DM problems.

LAWYERS VS ENGINEERS: GROUP 1

- A battery of tests/interviews with 30 engineers and 70 lawyers
- This resulted in thumbnail descriptions of these 30 engineers and 70 lawyers
- Two of these descriptions are now selected at random.
- Indicate your probability that the person is an engineer, on a scale from 0 to 100

JACK

- Jack is a 45-year old man. He is married and has four children. He is generally conservative, careful and ambitious. He shows no interest in political and social issues and spends most of his free time on his many hobbies which include home carpentry, sailing, and mathematical puzzles.
- What is the probability that Jack is an engineer (0-100)?

LAWYERS VS ENGINEERS: GROUP 2

- A battery of tests/interviews with 70 engineers and 30 lawyers
- This resulted in thumbnail descriptions of these 70 engineers and 30 lawyers
- Two of these descriptions are now selected at random.
- Indicate your probability that the person is an engineer, on a scale from 0 to 100

DICK

- Dick is a 30-year-old man. He is married with no children. A man of high ability and high motivation, he promises to be quite successful in his field. He is well liked by his colleagues.
- What is the probability that Dick is an engineer (0-100)?

IN AN EPIDEMIC, WHICH OF THE FOLLOWING TWO CHOICES DO YOU PREFER?

- CHOICE 3:
 - 8% chance of losing 100 lives
- CHOICE 4:
 - 10% chance of losing 75 lives

DISEASE X PROBLEM

1 out of every 1000 Americans has disease X. A test has been developed to detect when a person has disease X. Every time the test is given to a person who has the disease, the test comes out positive. But sometimes the test also comes out positive when it is given to a person who is completely healthy. Specifically, out of every 1000 people who are perfectly healthy, 50 of them test positive for the disease.

Imagine that we have assembled a random sample of 1000 Americans. Given the information given above: *How many people who test positive for the disease will actually have the disease? __ out of __*

YOU HAVE AGAIN 5 SECONDS!!!!

- Estimate the value of the following product:
- $8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$

WHY SOME OF THESE PROBLEMS?

- Philosophical tradition: People are rational!
Not always: tired, drunk, ...
- Tversky & Kahneman (and others):
Without fatigue, drugs, strong emotions, people make judgements that systematically violate rational norms.
- Findings are robust, but debate over what these findings indicate about the intrinsic rationality of ordinary people.
 - Pessimistic interpretation = bleak implications
 - People lack the underlying competence
 - However, two-systems, fast and frugal heuristics, ...